HisaabAI – MVP Scope

Problem statement: Small wholesale business owners in Bengaluru spend 1–2 hours every day manually reconciling handwritten ledgers against payments and invoices, because there's no quick way to see who owes what, who's overdue, or which products are actually selling.

Oct 21, 2026 · @Atharva

1. Problem

Small wholesale business owners lose 1–3 hours a day reconciling handwritten ledgers against cash, credit and stock — in their own words:

> "Every night after shop close, 3 hours manual work — checking cash sale, stock, who took udhaar, matching debtor ledger. And when I was not in shop, I had no idea if staff created / edited / deleted a bill." — wholesale dryfruits/spices shop owner, India

> "I'm new to business, and I've recently taken over my dad's small hardware retail store... I've used Loyverse and Google Sheets from the past few months." — hardware retail store owner

> "...the wholesalers who supply people like her — flour, boxes, packaging — are in the same boat. Most have zero presence online. No catalogue, just phone calls and a notebook of their own." — founder describing Indian bakery-supply wholesalers

GAP – needs research: none of these accounts is confirmed to be in Bengaluru or to sell specifically on credit to repeat shopkeeper customers — all three are adjacent personas, not exact matches.

2. Current State

1. Shop runs through the day — cash sales and credit ("udhaar") sales happen informally.
2. After closing, owner checks the day's cash sale total.
3. Owner checks physical stock.
4. Owner checks who took goods on credit that day.

5. Owner manually matches all of this against the debtor ledger.

"Every night after shop close, 3 hours manual work — checking cash sale, stock, who took udhaar, matching debtor ledger."

Tools people mention using today:

Tool	Evidence
Physical notebook / manual register	Khatabook review 4994375 · r/indianstartups
Excel / Google Sheets	r/BusinessPH · Vyapar review 13042620 · Vyapar review 12587292
Dedicated apps once adopted (Khatabook, Vyapar, Peddlr, Loyverse)	recurring across the Reddit and G2 datasets

3. Opportunity

Metric	Value	Source
Frequency	Daily ("every night after shop close")	reddit.com/r/Indian_Business
Time per task (wholesaler-specific)	3 hrs/day before, ~10–15 min/day after adopting a POS	same source
Time per task (generic reconciliation, not wholesale-specific)	20–40 hrs/month manual reconciliation	kosh.ai
Time saved via automation (generic)	20 hrs/month → under 5 hrs/month	kosh.ai
People/businesses affected (broad proxy)	60–70M SMBs in India, ~30% of GDP	linkedin.com/Bala Srinivasa
Market growth (broad proxy)	India retail & wholesale market, 7.8% CAGR, 2026–2032	6wresearch.com

GAP – needs research: no Bengaluru-specific figure exists in the data; the SMB-count and CAGR numbers are broad proxies, not specific to wholesalers selling on credit; the only wholesaler-specific time figure (3 hrs/day) is one anecdote, not a survey.

4. Solution

#	Step	Disappears as manual work?
1	Owner/staff photographs the handwritten ledger page or a receipt on their phone	Changes form, doesn't disappear
2	System extracts structured data from the photo (customer, item, amount, cash/credit, date)	Disappears — replaces "checking cash sale"
3	System matches the entry against the existing customer ledger, flags anything ambiguous	Disappears — replaces "matching debtor ledger"
4	System computes each customer's running balance and flags who is overdue	Disappears — replaces "checking who took credit"
5	System drafts a contextual reminder message per overdue customer	New capability — didn't exist before
6	Owner gets a daily dashboard: cash vs. credit split, overdue list, top/bottom-selling products	Disappears — replaces the 3-hour nightly reconciliation ritual with a ~5–10 min review
7	Owner approves/edits the reminder batch, then it sends	Stays human — deliberate checkpoint
8	System logs sent reminders and updates the ledger once payment is confirmed	Disappears — replaces manual bookkeeping of who paid, when

Net: steps 2–4 and 8 of the old nightly ritual (reddit.com/r/Indian_Business/comments/1wlk9oq) are fully automated; only capture (1) and approval (7) remain human.

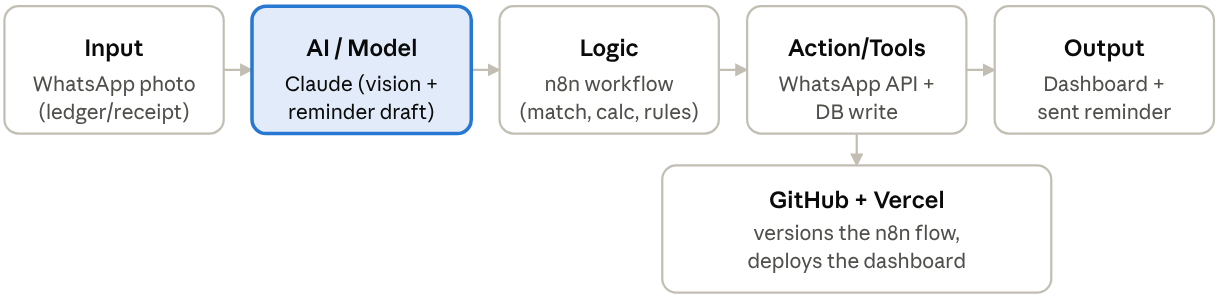
5. AI Thinking

Step	AI or Human/Logic?	Why
1. Capture photo	Human	Data originates on paper in someone's hand — the capture action can't be removed.
2. Extract structured data from image	AI	Handwriting varies per owner, mixes Kannada/Hindi/English, uses abbreviations — a fixed OCR script breaks on messy, varied input; a vision-language model generalizes.
3. Match entry to existing customer record	Logic, with AI fallback	Exact-ID matching is deterministic (a script is correct and cheaper); AI (fuzzy match) only kicks in for misspelled or ambiguous names.
4. Compute balance & overdue flag	Logic (script)	Pure arithmetic and date comparison against a clear rule — no ambiguity, so an LLM adds cost and hallucination risk for no benefit.
5. Draft reminder message	AI	Tone depends on context (VIP repeat customer vs. new customer vs. 60 days overdue) — hardcoding every combination into templates doesn't scale; an LLM composes it per case.
6. Daily dashboard numbers	Logic (script)	Sums, rankings and aggregates — deterministic, auditable math. Reviewers already distrust tools that can't show their work (Vyapar review 12284137).

Step	AI or Human/Logic?	Why
7. Approve & send	Human	This is exactly the control G2 reviewers said is missing elsewhere — "can't track who changed what and when" (Vyapar review 12284137). A human send-gate protects real relationships and catches AI mistakes before money is at stake.
8. Log payment / update ledger from receipt photo	AI (extraction) + Logic (state update)	Same reasoning as step 2 for the photo; the ledger state transition itself is deterministic.

6. Architecture

Photo in, dashboard and reminder out



architecture · Input → AI/model → Logic → Action/tools → Output

Claude is the only AI step — extracts structured data from the ledger/receipt photo, and drafts the reminder text. **n8n** orchestrates everything deterministic: matching, balance/overdue calculation, business rules (skip if paid <24h, cap reminder frequency), and calls out to WhatsApp and the database — this is where the audit trail G2 reviewers said was missing (Vyapar review 12284137) gets built in by default. **GitHub** version-controls the n8n workflow and the dashboard code. **Vercel** hosts the owner-facing dashboard with CI/CD from GitHub.

7. Impact

Primary estimate (wholesaler-specific, single-source anecdote):

- Before: 3 hrs/day → After: ~15 min/day (0.25 hrs/day) — reddit.com/r/Indian_Business
- Daily savings: 3 − 0.25 = **2.75 hrs/day**
- Assumption (not in the data): 6-day working week, typical for small Indian retail/wholesale
- Weekly savings: 2.75 × 6 = **~16.5 hours/week per user**

Conservative sanity check (generic reconciliation benchmark, not wholesale-specific):

- Before: 20–40 hrs/month → After: under 5 hrs/month — kosh.ai · kosh.ai
- Monthly savings: 15–35 hrs/month ÷ 4.3 weeks = **~3.5–8 hours/week**

Range to use in a pitch: 3.5–16.5 hrs/week per user — low end from a generic accounting-software benchmark (safer), high end from a direct wholesaler account (less validated).

Quality: reduces data-entry errors and missed follow-ups (the core pain in the problem statement) by making balance/overdue status computed, not manually tallied — no dataset metric validates an error-rate reduction; mark as GAP – needs research.

Scale: 60–70M SMBs in India is the only population proxy in the data (linkedin.com) — not wholesale- or Bengaluru-specific, so treat as a ceiling, not a target.

GAP – needs research: get 5–10 real Bengaluru wholesalers to time their own nightly reconciliation before quoting a number with confidence.

8. Limitations

#	Failure case	What a human checks
1	OCR misreads a handwritten amount or name (e.g. "500" read as "5000," or two similar customer names merged)	Owner reviews the extracted summary against the original photo before it's committed, especially for high-value entries or new/unrecognized names
2	AI drafts a reminder with the wrong tone, amount, or timing (e.g. curt message to a large repeat customer, or a reminder sent after an unrecorded cash payment)	Every reminder batch sits in an approval queue (step 7) — owner edits or skips individual messages before anything sends

#	Failure case	What a human checks
3	Duplicate or mismatched ledger entries (same transaction photographed twice, or a payment fuzzy-matched to the wrong invoice via a similar name)	Any match below a confidence threshold routes to a "needs review" queue on the dashboard instead of auto-resolving silently

Existing tools (Khatabook, Vyapar) already handle the core "who owes what / who's overdue" ask well — reviewers praise their reminder and low-stock-alert features, not complain about them (Khatabook review 6485190). The gaps they do name — reporting depth, audit trail, integrations — are what this MVP should target first; see the companion research doc's Limitations section for the full review evidence.

What we'd build next (v2+): OCR confidence scoring UI, multi-language handwriting support (Kannada/Hindi/English mixed), a change-log/audit trail for staff edits (the gap Vyapar reviewers flagged), and product-level sales analytics (unverified as a real pain point in this dataset — GAP – needs research).

9. MVP Scope

IN (build by Thu 12:30 PM) — the ONE core user journey: *owner photographs a ledger entry → sees it digitized with an updated balance → gets a drafted reminder to approve and send.*

1. WhatsApp photo capture → Claude extracts structured transaction data
2. n8n computes running balance per customer + flags overdue
3. Claude drafts a reminder; owner approves in a simple dashboard before it sends

OUT (mention, don't build): product/stock-performance analytics (unverified pain point — see Limitations), multi-language OCR beyond English/Hindi/Kannada basics, e-commerce/tax-software integrations, multi-device/multi-staff permissions.

LATER (v2): audit trail of staff edits, OCR confidence UI, automatic payment-received detection from a photographed receipt, product-level sales insights.

Magic moment: the owner photographs their handwritten ledger page at closing and, within seconds, sees exactly who owes what and a ready-to-send reminder — the 3-hour nightly "hisaab" becomes a single approval tap.

Tech stack:

Tool	Role
Claude	Extracts structured data from ledger/receipt photos; drafts contextual reminder messages
n8n	Orchestrates the pipeline: matching, balance/overdue calculation, business rules, calls to WhatsApp and the database
GitHub	Version-controls the n8n workflow and dashboard code
Vercel	Hosts the owner-facing dashboard with CI/CD from GitHub

Sources

Reddit (trudax/reddit-scraper-lite, 38 items):

- r/Indian_Business — "I built a POS for my own wholesale dryfruits/spices shop"
- r/BusinessPH — "Are there other apps like Peddlr?"
- r/indianstartups — "Built software for India's bakers (and the wholesalers who supply them)"

Google Search (apify/google-search-scraper, 3 queries, 27 organic results):

- kosh.ai — Automated Reconciliation Systems
- kosh.ai — The Hidden Costs of Manual Reconciliation
- linkedin.com — India's Small Business Market Paradox
- 6wresearch.com — India Retail And Wholesale Market

G2 (automation-lab/g2-scraper, 10 Khatabook + 20 Vyapar reviews):

- Khatabook review 4994375
- Khatabook review 7402849
- Khatabook review 6537176
- Khatabook review 6485190
- Vyapar review 12284137
- Vyapar review 13042620
- Vyapar review 12587292

- Vyapar review 13397992
- Vyapar review 13384261
- Vyapar review 12128442